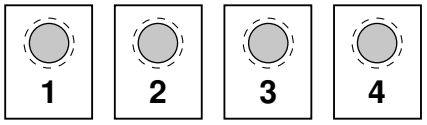
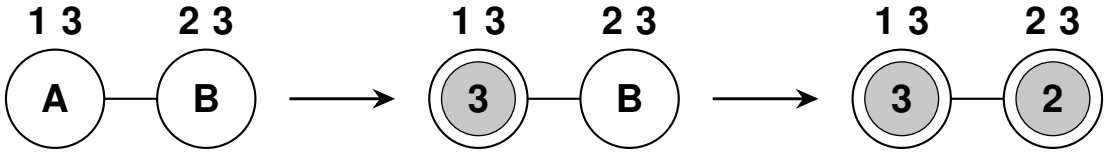
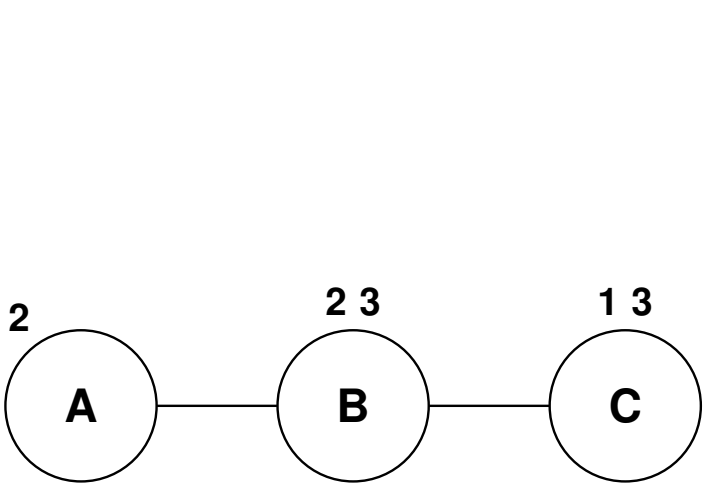
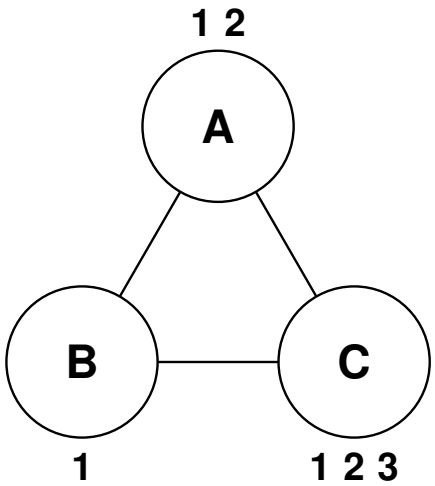
The headers show which colour goes with each number. Put one counter on each circle. Two circles joined by a line must not have the same number. The numbers next to a circle show which counters it may have.

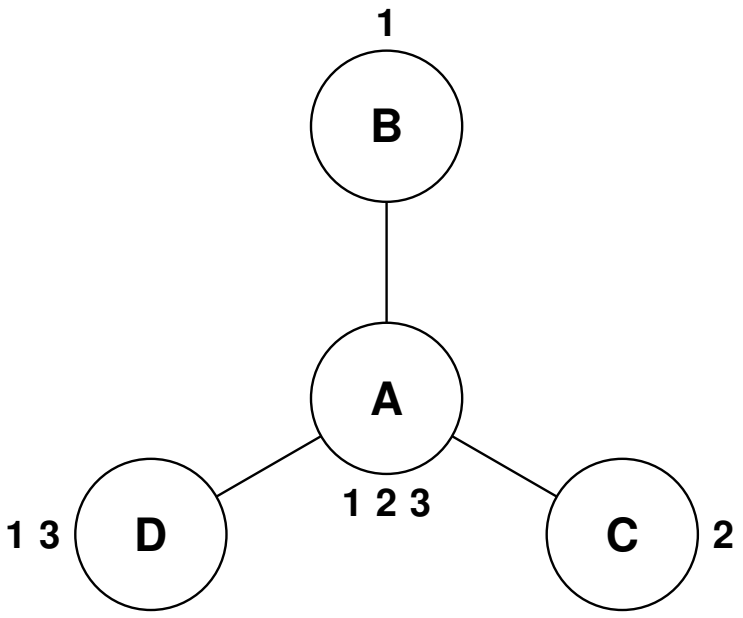


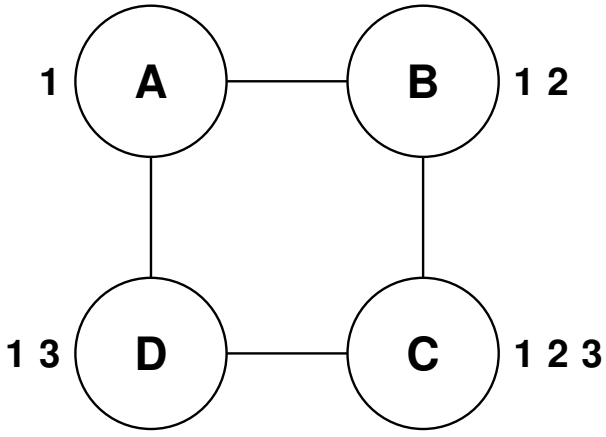


Problem 1: Put a counter on every circle.

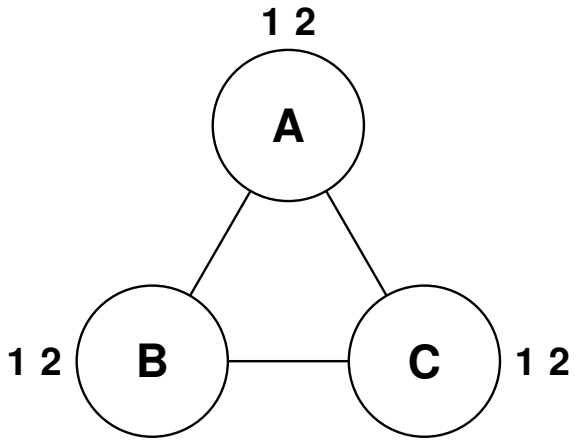


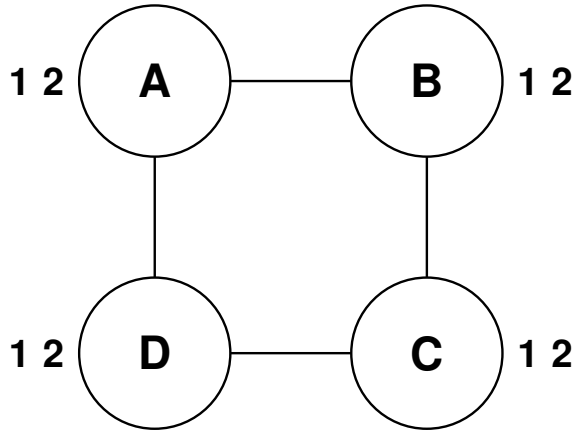


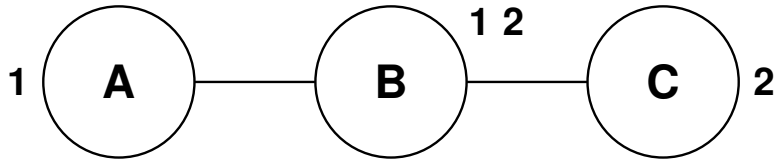


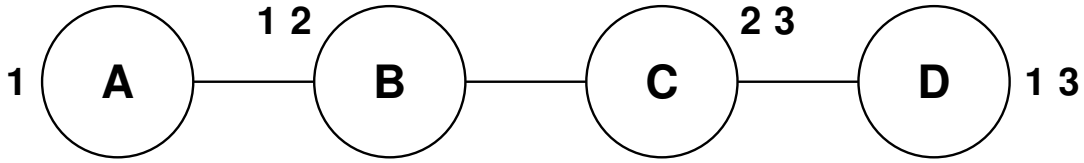


Problem 2: Some of these can't be done. Find them and show why.

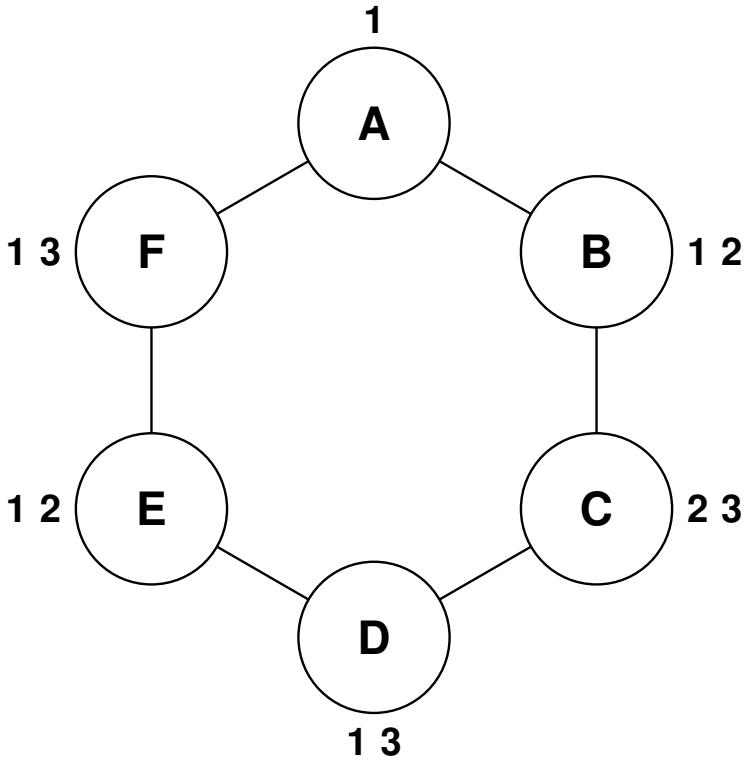


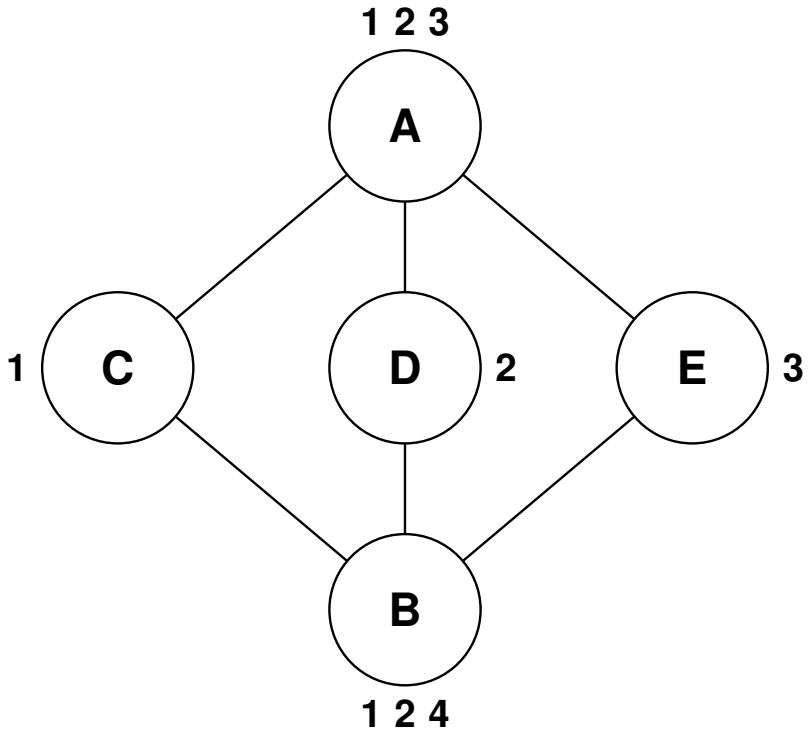




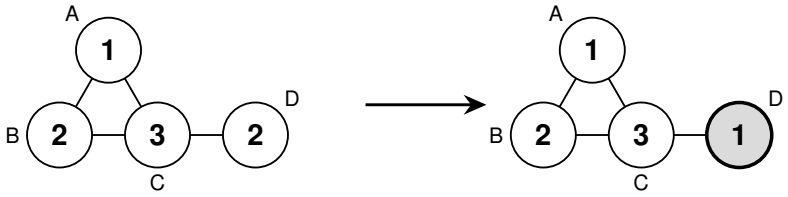


Problem 3: Put a counter on every circle, or show why it can't be done.

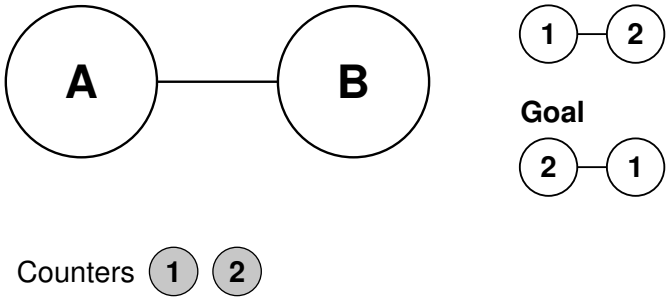




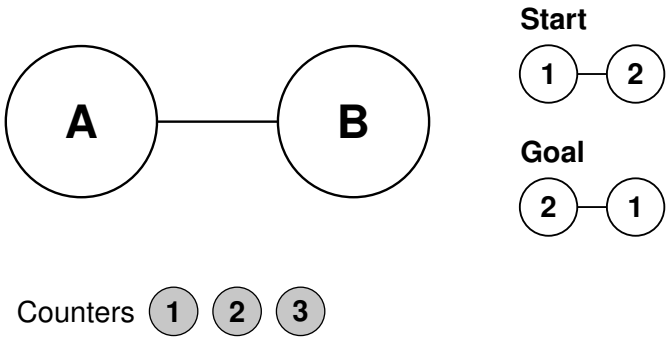
A move takes one counter off a circle and puts a counter with a different number there. After every move, circles joined by a line must still have different numbers.



Problem 4: Set up Start. Can you change it into Goal one move at a time, using only the counters shown?

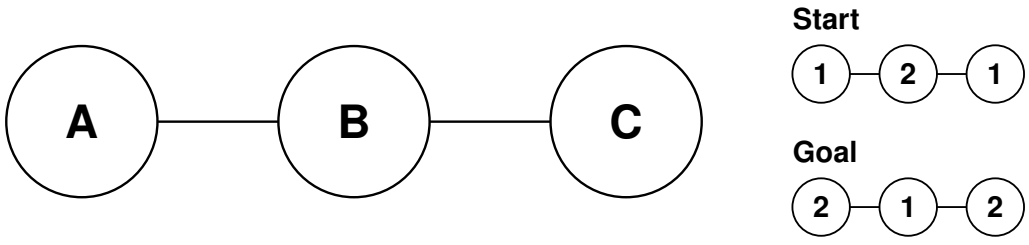


Counters 1 2



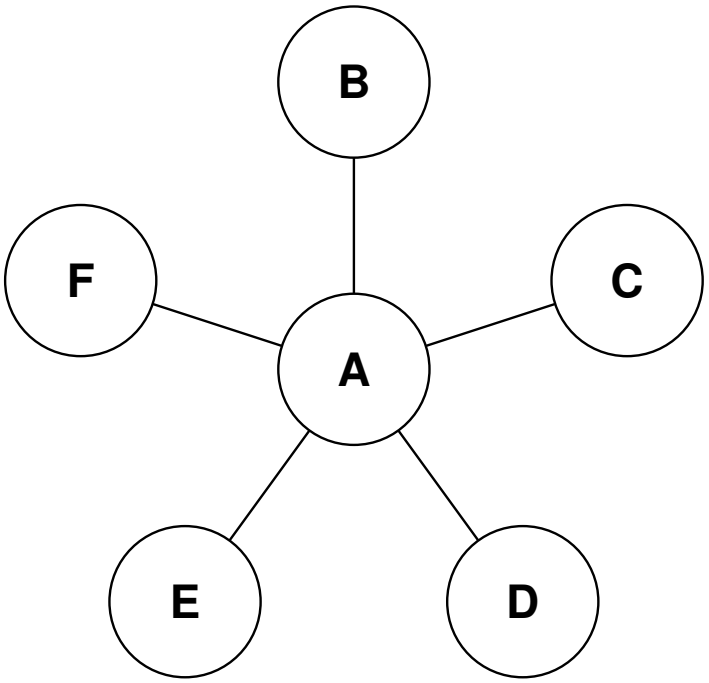
Counters 1 2 3

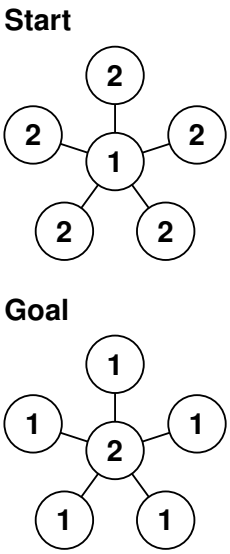
Problem 5: Set up Start, then change it into Goal in as few moves as you can.



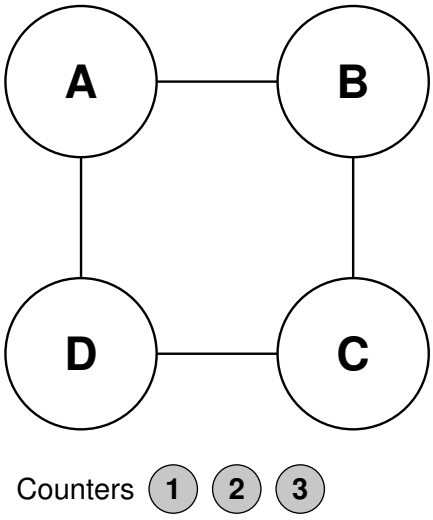
Counters 1 2 3

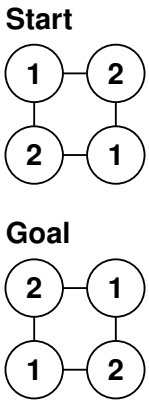
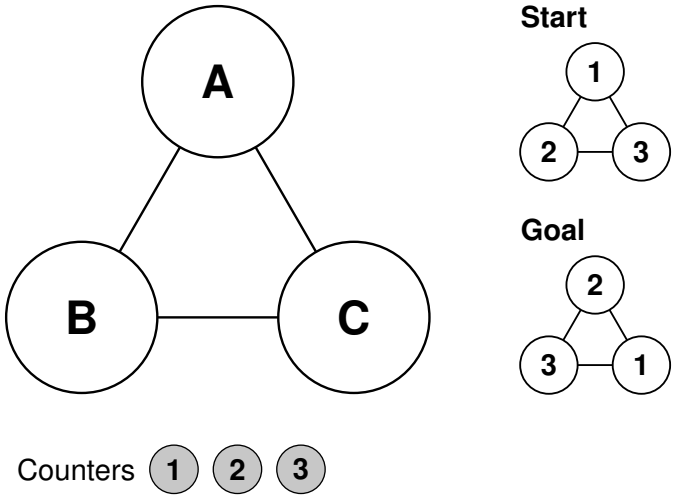
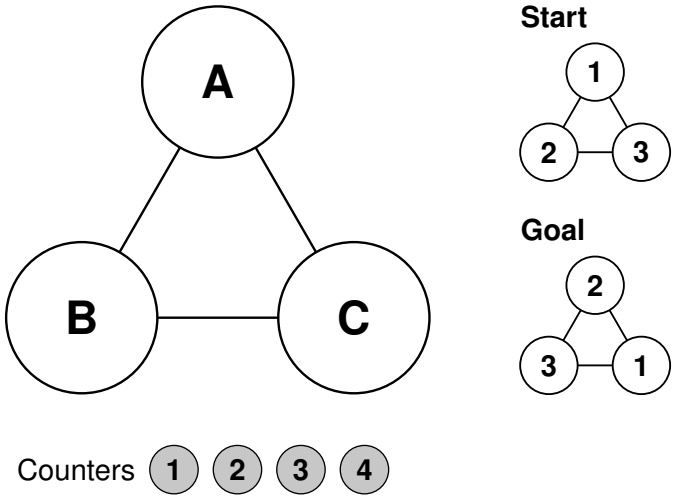
Problem 6: Set up Start, then change it into Goal in as few moves as you can.





Problem 7: Set up Start, then change it into Goal in as few moves as you can.





Problem 8: Set up Start. Can you change it into Goal one move at a time, using only the counters shown?